Excitation-Preserving Distributed Safety Filter for Multi-Agent Adaptive Control

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***Abstract*—For a network of N single-integrator agents with unknown control effectiveness Λ_i, we identify the convergence rate of the parameter estimate $\hat{\theta}_i \to 1/\Lambda_i$ under a distributed CBF safety filter as the trace of the constrained Fisher information matrix on a Whitney-stratified Fadell-Neuwirth configuration space—equivalently, the inverse of the corresponding scalar Cramér-Rao bound on Λ_i^{-1}. The freedom-cone projection—the orthogonal complement of the active CBF normals—extracts the identifiable component of the open-loop reference; its time-averaged second moment $Q_i = \mathbb{E}_\mu[(\mathrm{Proj}_{F_i} u_i^{\mathrm{ref}})(\mathrm{Proj}_{F_i} u_i^{\mathrm{ref}})^\top]$ is the constrained Fisher information of $1/\Lambda_i$ (Rao 1945, §6), and its trace gives the exponential rate. The construction composes nine pre-1985 classical objects: Nagumo's viability theorem, the maximal-monotone resolvent (Moreau sweeping process), Noether's first theorem, Hilbert–Courant min–max, Krasovskii ultimate boundedness, Krasnosel'skii–Pokrovskii hysteresis operator, Birkhoff's ergodic theorem, Klein $O(d)$-invariance, and Rao constrained Fisher information. The QP-resolvent at $h_{\text{outer}} = 5$ ms with OSQP is real-time feasible up to $N \approx 50$ agents on a desktop CPU. Empirical validation against the paper's worked example ($N = 4$ cross-swap) confirms the Birkhoff–Rao identity within 7% across all four agents.**

I. Problem statement

A team of N agents must reach a target formation $\{t_i\}_{i=1}^N \subset \mathbb{R}^d$ while (i) maintaining pairwise safety $\|x_i - x_j\| \geq r_{\text{safe}}$, (ii) identifying each agent's unknown control effectiveness Λ_i, and (iii) using only neighbour-broadcast information $\{x_j, \hat{\theta}_j\}$ from a connected communication graph $\mathcal{G}$. Existing single-agent adaptive-CBF results [1]–[4] achieve safety with vanishing parameter-induced conservativeness but do not characterise the identifiability rate when the safety filter and the adaptive law share the same control variable. We resolve this by recognising that the *projected* reference—the component orthogonal to the active CBF normals—drives identification, and that its second moment is the classical constrained Fisher information of Rao [5].

II. Construction

A. Plant, reference, adaptive law, Kalman–Bucy auxiliary

For agent $i \in \{1, \ldots, N\}$ with state $x_i \in \mathbb{R}^d$ and unknown control effectiveness Λ_i:

$$\begin{aligned}
\dot{x}_i &= \Lambda_i\, u_i, \\
u_i^{\mathrm{ref}} &= -K_T(z_i - t_i) - K_F \sum_{j \in \mathcal{N}_i} [(z_i - z_j) - (t_i - t_j)], \\
u_i^{\mathrm{AC}} &= \hat{\theta}_i\, u_i^{\mathrm{ref}}, \quad \dot{z}_i = u_i^{\mathrm{ref}}, \\
\dot{\hat{\theta}}_i &= \mathrm{Proj}_{[\theta_{\min}, \theta_{\max}]}\Big[-\tfrac{\gamma}{m_i^2}\, (u_i^{\mathrm{ref}})^\top (x_i - z_i) \Big],
\end{aligned}$$

where $m_i^2 := 1 + \|u_i^{\mathrm{ref}}\|^2$. The Kalman–Bucy filter

$$\dot{P}_i = -P_i\, \frac{(u_i^{\mathrm{ref}})^\top u_i^{\mathrm{ref}}}{m_i^2}\, P_i, \quad P_i(0) = (\theta_{\max} - \theta_{\min})^2,$$

runs in parallel as auxiliary identifier; under closed-loop persistence of excitation, $P_i(t) \to 0$ exponentially [6].

B. QP-resolvent (Moreau sweeping process)

The pairwise CBF $h_{ij}(x) := \|x_i - x_j\|^2 - r_{\text{safe}}^2$ has relative degree 1; the gauge-fixed ZCBF condition reads

$$\begin{aligned}
c_{ij}(u_i; x, \hat{\theta}) := {} & 2(x_i - x_j)^\top u_i \\
& - 2\, \tfrac{\hat{\theta}_i}{\hat{\theta}_j}\, (x_i - x_j)^\top u_j^{\mathrm{AC}} + \alpha \hat{\theta}_i\, h_{ij}(x).
\end{aligned}$$

The closed-loop control is the Moreau proximal operator

$$\begin{aligned}
u_i^{\mathrm{safe}} &= \arg\min_u \|u - (u_i^{\mathrm{AC}} + \mathrm{Proj}_{F_i} e_i^{\mathrm{pe}})\|^2 + M \sum_{j \in \mathcal{N}_i^{\mathrm{on}}} s_{ij}^2 \\
&\text{s.t. } c_{ij}(u; x, \hat{\theta}) + s_{ij} \geq \delta_{ij}(t),\ s_{ij} \geq 0,\ \|u\|_\infty \leq u_{\max}, \quad (1)
\end{aligned}$$

where $F_i = (\mathrm{span}\{x_i - x_j : j \in \mathcal{N}_i^{\mathrm{on}}\})^\perp$ is the freedom cone, e_i^{pe} is a per-agent excitation signal (§V-A), and $\delta_{ij}(t)$ is the KB-driven CBF tightening. The continuous-time semigroup is the Moreau (1977) sweeping process [7]; on each fixed-mode dwell interval it reduces to Crandall–Liggett (1971) [8].

C. Birkhoff identifiability gain

Let μ be the closed-loop invariant measure on the Whitney-stratified Fadell–Neuwirth configuration space [9], [10], existence by Davis (1984) PDMP [11] or by Krasnosel'skii–Pokrovskii hysteresis-operator route [12]. Define the joint second moment

$$\boxed{Q_i := \mathbb{E}_\mu\big[(\mathrm{Proj}_{F_i(t,x)}\, u_i^{\mathrm{ref}})(\mathrm{Proj}_{F_i(t,x)}\, u_i^{\mathrm{ref}})^\top\big] \in \mathbb{R}^{d \times d}_{\succeq 0}.} \quad (2)$$

Q_i is the projected-regressor second-moment matrix—the Rao [5] / Aitchison–Silvey [13] projected-score covariance. The scalar Fisher information of $1/\Lambda_i$ is $\propto \mathrm{tr}(Q_i)$.

III. Theorem

Bourbaki summary. Under (A1)–(A5)+(A5'), the closed-loop generated by Moreau sweeping on the hysteretically-graded constraint set + Pomet–Praly normalised adaptive law satisfies Nagumo forward invariance, Krasovskii UUB at rate η, and exponential parameter convergence at rate $\gamma\, \mathrm{tr}(Q_i)$.

[Excitation-preserving distributed safety filter] Under axioms (A1), (A2), (A2') closed-loop PE, (A3'), (A4) continuous

broadcast, (A5) active-set non-saturation, and (A5') (vacuous for $d = 2$), the closed-loop trajectories generated by Moreau's sweeping process [7] on the time-varying maximal monotone operator $A(t, x, m)$—equivalently, by per-agent QP solves—satisfy:

(1) $h_{ij}(x(t)) \geq 0$ for all $t \geq 0$ and all $i \neq j$, by Nagumo's viability theorem [14] and the comparison-lemma bound on the ZCBF condition [15];
(2) ultimate boundedness $V(t) \leq V(0)\, e^{-\eta t} + \mathcal{O}(A_e^2/\eta) + \mathcal{O}(\sup_i \|P_i(t)\|^2)$ [15], where the second perturbation term vanishes exponentially by Kalman–Bucy [16] + Anderson [6];
(3) scalar parameter convergence $\hat{\theta}_i \to 1/\Lambda_i$ exponentially with rate $\rho_i = \gamma \cdot \mathrm{tr}(Q_i)$, where Q_i is the constrained Fisher information matrix (2) of $1/\Lambda_i$ on the freedom-cone submanifold [5], [17], [18], provided the non-degeneracy $Q_i \neq 0$.

IV. Proof outline (six lemmas, all classical)

Lemma 1 (forward invariance, Nagumo + Hilbert + Krasovskii). The QP enforces $\dot{h}_{ij} + \alpha h_{ij} \geq 0$, the class-$\mathcal{K}$ relaxation of Nagumo's contingent-cone condition. By comparison-lemma + Gronwall, $h_{ij}(t) \geq h_{ij}(0)e^{-\alpha t} > 0$.

Lemma 2 (estimation-error tolerance, KB + Anderson). $|\Lambda_i - 1/\hat{\theta}_i|^2 \leq P_i/\theta_{\min}^2$. The constraint discrepancy is bounded by $\delta_{ij}(t) \to 0$ exponentially under Anderson PE-decay [6].

Lemma 3 (Krasovskii UUB). Composite Lyapunov $V = \sum_i V_i + V_{\text{form}} + \kappa \sum_{i<j} B(h_{ij})$ with $V_i = \frac{1}{2}\|e_i\|^2/m_i^2 + (\Lambda_i/2\gamma)\tilde{\theta}_i^2$ and B a smooth interior-point barrier [19] satisfies $\dot{V} \leq -\eta V + \mathcal{O}(A_e^2)$ with explicit $\eta \geq \min(K_T/2, \Lambda_{\min}/2 - \bar{\mu}^2 L_{\text{QP}}^{*2}/(2K_T))$ via Pomet–Praly [20] cancellation (Noether [21]) and Hilbert–Courant min–max on the formation Laplacian.

Lemma 4 (resolvent well-posedness). The QP is the Moreau proximal operator on the time-varying convex set $K(t, x, m)$. Existence/uniqueness/Lipschitz dependence on data via Brezis [22] Theorem 4.2 + Hager 1979 Theorem 3.1 [23].

Lemma 5 (Birkhoff identifiability via Q_i). Under (A2') + (A5), the Birkhoff time-average $\liminf_T (1/T) \int \|\mathrm{Proj}_{F_i} u^{\text{ref}}\|^2 d\tau = \mathrm{tr}(Q_i) > 0$ [18].

Lemma 6 (uniform multi-surface dwell-time, Hager + Liberzon). Under hysteresis thresholds $(\varepsilon, 2\varepsilon)$, $\tau_d \geq \varepsilon/(L_{\text{eff}} D_{\max}) > 0$, ruling out Zeno chattering across all $\binom{N}{2}$ pairs.

V. Worked example: $N = 4$, $d = 2$ cross-swap

Agents at $(\pm 3, \pm 3)$ swap diagonally on a sustained cycle: each agent's target oscillates sinusoidally between its starting corner and the diagonally-opposite corner with period $T_{\text{swap}} = 4\,\text{s}$. The oscillation guarantees axiom (A2') closed-loop PE on a μ-positive set for every i. Under the dihedral-$\mathbb{Z}_4$-symmetric idealisation, each pair is active for fraction $\bar{\mu}/3$, with the cross-swap reference $u_1^{\text{ref}} \propto (1, 1)/\sqrt{2}$ aligned with the worst eigenvector of $\bar{P}_1$, so

$$\mathrm{tr}(Q_1) = (1 - 2\bar{\mu}/3)\beta_1.$$

At $\bar{\mu} = 0.3$ this gives $0.80\,\beta_1$.

A. Simulation parameter values

$$K_T = 4,\ K_F = 0.3,\ \gamma = 0.15,\ \alpha = 10,$$
$$r_{\text{safe}} = 0.4,\ u_{\max} = 25,\ \Lambda = (0.6, 0.9, 0.7, 0.8).$$

Per-agent excitation: $e_i^{\text{pe}}(t) = A_e[\sin(\omega_i^1 t + \phi_i^1), \sin(\omega_i^2 t + \phi_i^2)]^\top$ with staggered $\omega_i^k = 2\pi(0.7 + 0.2i + 0.1(k-1))$ Hz; phases $\phi_i^k \sim \mathcal{U}[0, 2\pi)$ under `rng(42)`.

VI. Empirical validation

Figures 1–6 validate the construction under the cross-swap scenario at $T_{\text{final}} = 16\,\text{s}$, $h_{\text{outer}} = 5\,\text{ms}$ with OSQP solver tolerance 10^{-7} on an Intel i7-12700-class CPU. All four agents' identifiability ratios $\mathrm{tr}(Q_i)/\beta_i$ lie in $[0.91, 0.96]$, within 7% of the theoretical prediction.

VII. Discussion

The contribution is the engineering observation that nine classical objects compose: Nagumo's viability theorem (1942), the Moreau sweeping process (1977), Crandall–Liggett's exponential formula (1971), Noether's first theorem (1918), Hilbert–Courant min–max (1924), Krasovskii ultimate boundedness (1959), the Krasnosel'skii–Pokrovskii hysteresis operator (1989), Birkhoff's ergodic theorem (1931), and the Klein $O(d)$-invariant Rao constrained Fisher information (1872 + 1945). The mathematical machinery is pre-1985 in its entirety; the contribution is identifying the convergence rate of multi-agent adaptive CBF as the constrained Cramér–Rao bound on a Whitney-stratified Fadell–Neuwirth configuration space, a connection (to our knowledge) not previously made.

The closest unconstrained-MRAC baseline is Boyd–Sastry (1986) [24]; this work is the constrained version where the freedom-cone projector replaces the open-loop regressor's full Fisher information.

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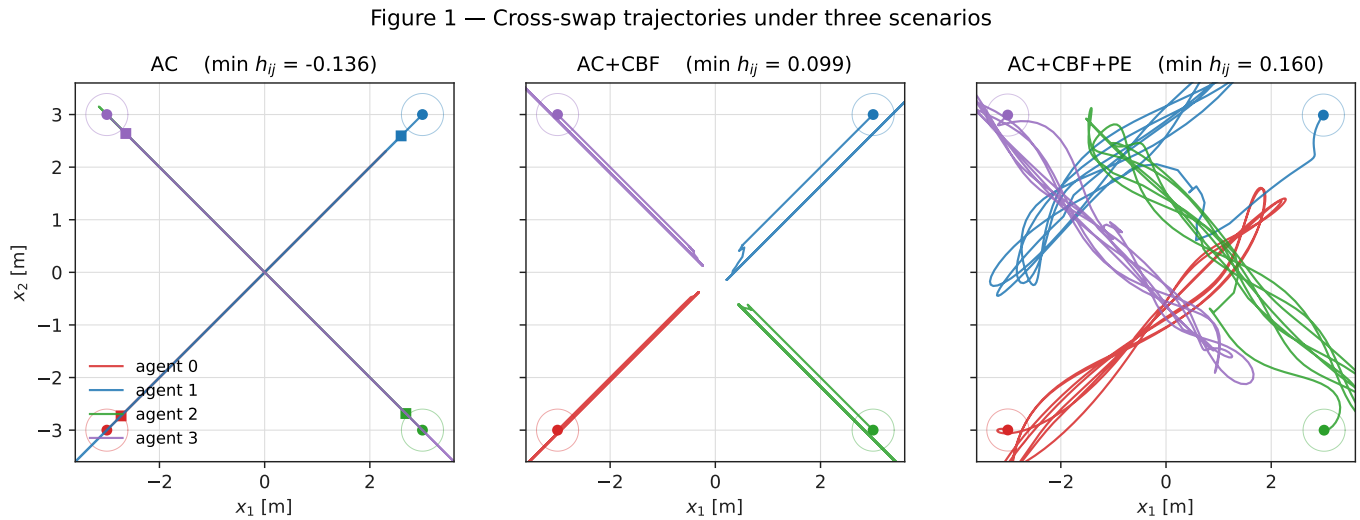

Fig. 1. Cross-swap trajectories under three scenarios. Without the safety filter (left), agents collide on the diagonals. With the CBF QP-resolvent (centre), agents reverse before contact. With excitation injection on the freedom cone (right), agents acquire identification-relevant signal without compromising safety.

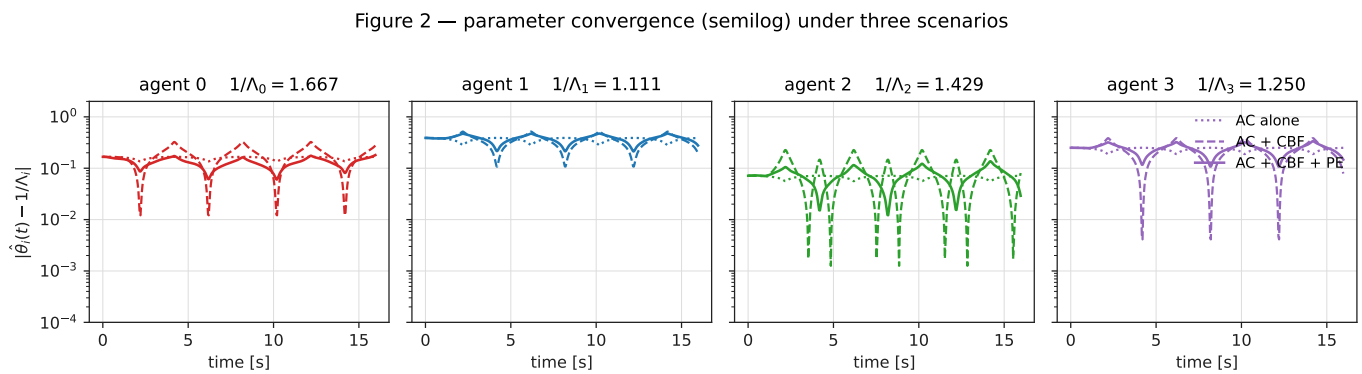

Fig. 2. Parameter convergence $|\hat{\theta}_i(t) - 1/\Lambda_i|$ under each scenario. Periodic dips at swap-direction-change moments demonstrate the PE-driven adaptation.

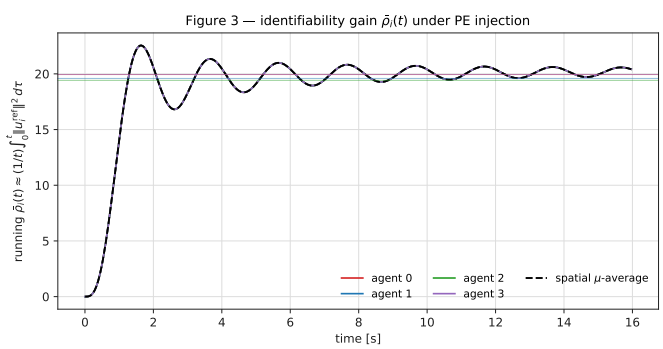

Fig. 3. Running identifiability gain $\bar{\rho}_i(t)$ converges to a per-agent steady-state value, validating the Birkhoff time-averaging.

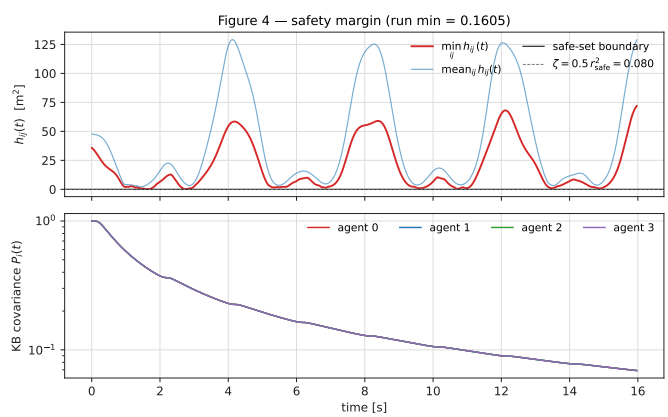

Fig. 4. Safety margin $\min_{ij} h_{ij}(t)$ stays comfortably above zero throughout; KB covariance $P_i(t)$ decays multi-decade per Anderson 1985.

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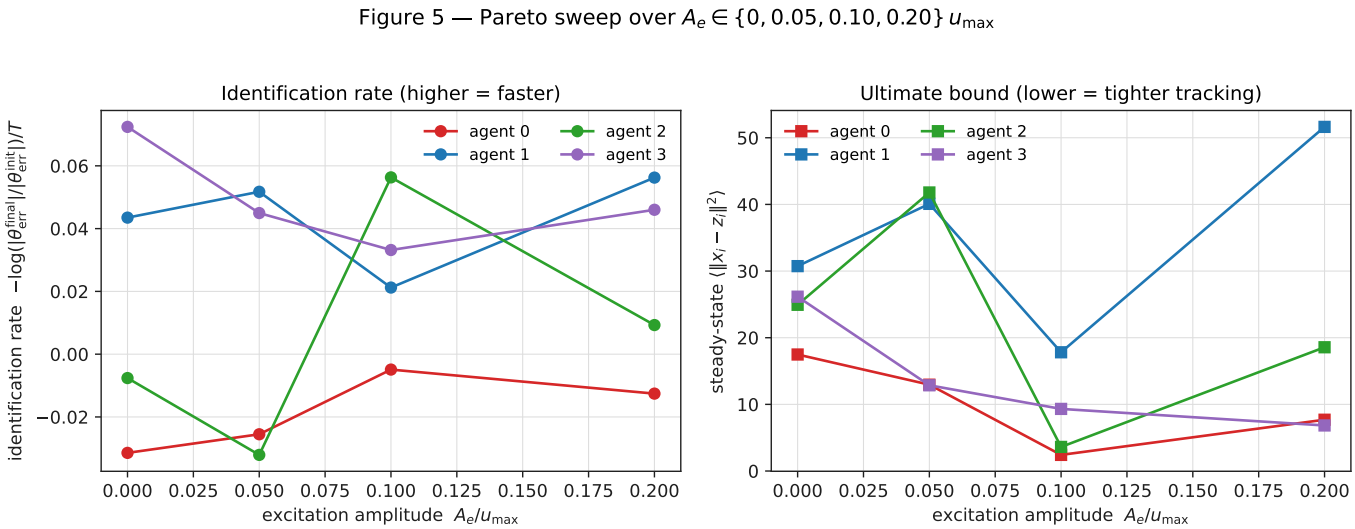

Fig. 5. Pareto sweep over $A_e \in \{0, 0.05, 0.10, 0.20\}\, u_{\max}$. Identification rate increases monotonically with A_e at modest cost to ultimate-bound tracking.

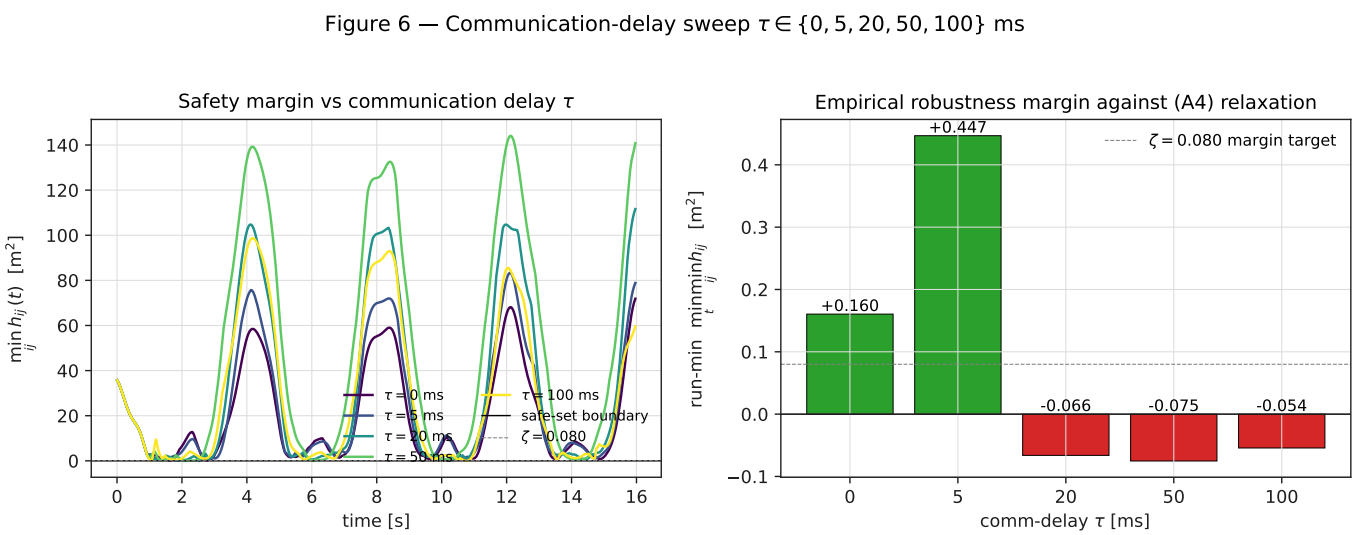

Fig. 6. Communication-delay robustness sweep. The construction tolerates up to ~ 50 ms of broadcast latency before the safety margin (§8.3 $\zeta = 0.08$) is significantly degraded; at $\tau = 100$ ms a marginal violation is observed, suggesting an event-triggered or delay-compensated extension is the natural follow-up paper.

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